

zero padding:- adding zeros to the size to solve your problem.

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Recurrent Neu. Net:-

Special class of neu. networks used for sequences.

- RNN is type of sequential model specifically designed to work on seq. data.

Hi my name is farhan → seq. data.

Simply means where seq. matters.

→ RNN → NLP

Why use RNN:-

Input

Output.

Hi - my name is farhan

0

I love ml

0

Pak won the match

1

- Problem is textual data may vary in size but ANN input size doesn't vary or change.

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- If u use ANN so even if u do zero padding but still u do a lot of unnecessary computation.
- Prediction problem.

Application:-

Google translator

Roadmap:-

Simple RNN $\rightarrow$ B.P RNN $\rightarrow$ LSTM $\rightarrow$

Types of RNN $\rightarrow$ Deep RNN $\rightarrow$ GRU

Bidirection RNN $\rightarrow$ Seq to seq models

How RNN works:-

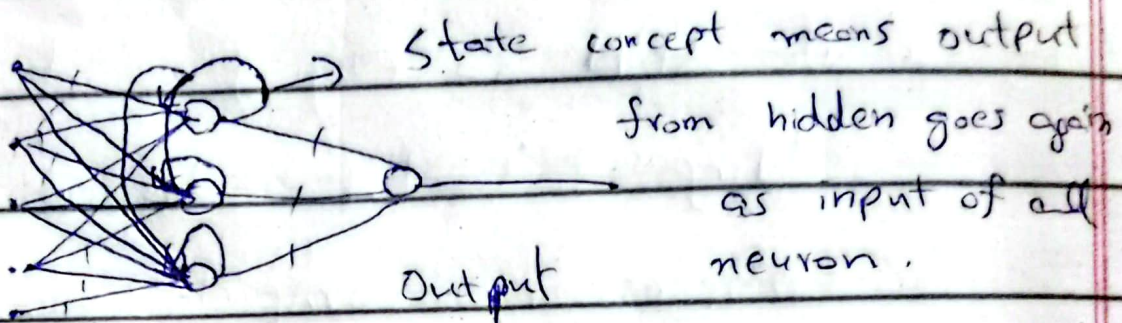
	Review	Sentiment
x_1	x_{11} movie x_{12} was x_{13} good	1
x_2	x_{21} movies x_{22} was x_{23} bad.	0
x_3	x_{31} movies x_{32} was x_{33} not x_{34} good.	0

RNN $\xrightarrow[\text{similar like}]{\text{very}}$ ANN $\xrightarrow[\text{differences}]{\text{2 major}}$ 1 is timestamp $\rightarrow$ state concept.

Timestamp means at a time only
1 word will proceed.

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Input hidden funct = Sigmoid as it is sentiment.

Now weights are

$$W = 5 \times 3 = 15 + 3 \times 3 = 9 + 3 = 27.$$

$$b = 3 \times 1 = 4$$

Forward propagation:-

Review Sentiment.

x_{11}, x_{12}, x_{13}

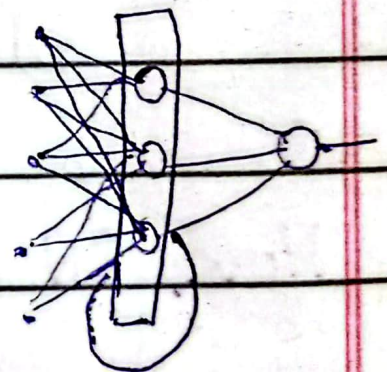
1

x_{21}, x_{22}, x_{23}

0

x_{31}, x_{32}, x_{33}

0



$x_{11} \rightarrow$ Vectors

$[1, 0, 0, 0]$

$b + \text{tanh} \rightarrow$
 $x_{11} \uparrow w_i$

O_1

$b_2 + \text{tanh} \rightarrow$
 $x_{12} \uparrow w_i$

O_2

for consistency pass

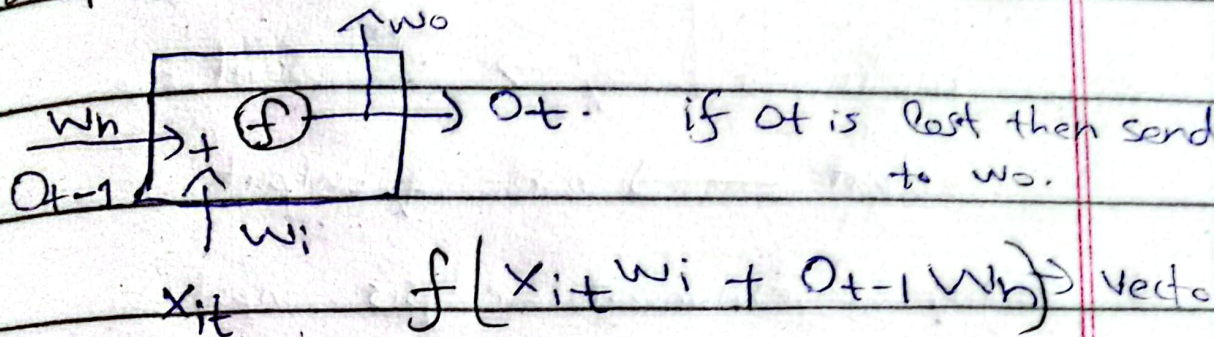
O_0 of $[1, 3]$

Multiclass $\rightarrow$ Regression
 $\downarrow$ $\downarrow$
 Softmax Linear

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Simplified representation:-



i = row no.

t = time step

x_{it}

Types:-

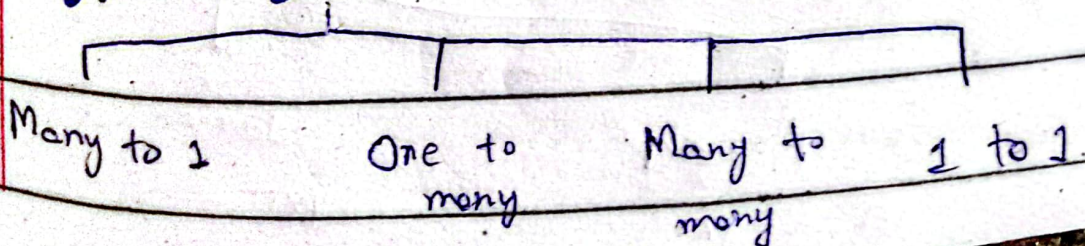
Integer-encoding:- Convert text to nos.

Embedding:- Convert integers to vectors

I love AI } I.E.
 1 2 3

love 2 $\rightarrow [0.3, 0.7, 0.1] \rightarrow$ Embedding

Types of RNN:-



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Many to 1:-

Input $\xrightarrow{\text{expect}}$ Seq, $\xrightarrow{\text{It should be}}$ Sent, charact, time series
output $\longrightarrow$ non-seq output $\longrightarrow$ int/num

e.g :- Sentiment analysis input $\rightarrow$ text $\rightarrow$ seq.

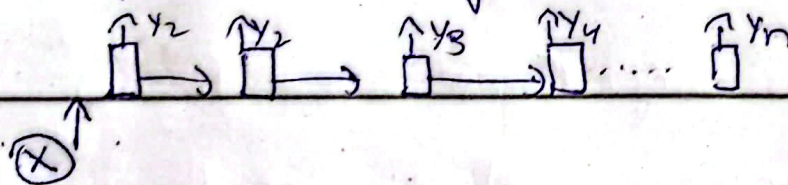
$\rightarrow$ rating prediction.

output $\rightarrow$ 1, 0 $\rightarrow$ scalar.

• 1 to Many:-

Input $\rightarrow$ normal non-sequential image, no

Output $\rightarrow$ Sequential. (like word, seq.) ^{could be}



e.g:- image $\rightarrow$ N.N $\rightarrow$ men playing cricket

• Many to Many:-

Input $\rightarrow$ Sequential $\rightarrow$ Seq to Seq models.

Output $\rightarrow$ " : : : "

Same length

M. + M

Variable length

M. + M.

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input seq = output seq. inp $\neq$ out

e.g.: POS tagging.

e.g.: google translate

- One to 1:-

input $\rightarrow$ non seq

output $\rightarrow$ " "

technically not RNN. So there

are 3 types 😊